

English
Medium



EDU TERIA

72nd BPSC

Prelims Online Batch

BIOLOGY

Topic wise
Notes

LECTURE

24

Plant Kingdom

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The Plant Kingdom, or **Kingdom Plantae**, consists of multicellular, eukaryotic organisms that are primarily **autotrophic**, meaning they produce their own food through photosynthesis.

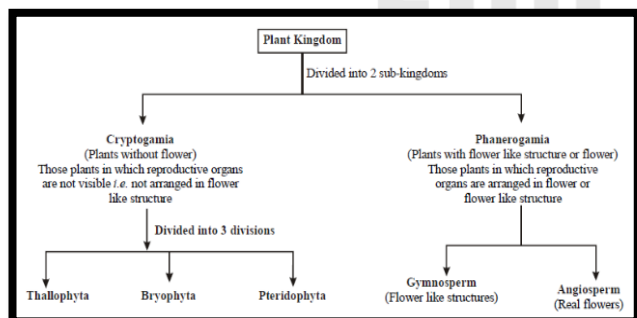
Facts related to Plant Kingdom

- All eukaryotic multicellular autotrophic organisms are kept in this kingdom.
- Branch deals with study of plants known as Botany.
- Father of Botany → Theophrastus Historia planterum.

A.W Eichler Plant classifications

A. W. EICHLER

- Eichler gave the first phylogenetic classification of plant kingdom.
- Eichler classified plant kingdom into two sub-kingdoms- cryptogamia and phanerogamia
- The classification of Eichler is very little phylogenetic.



Algae

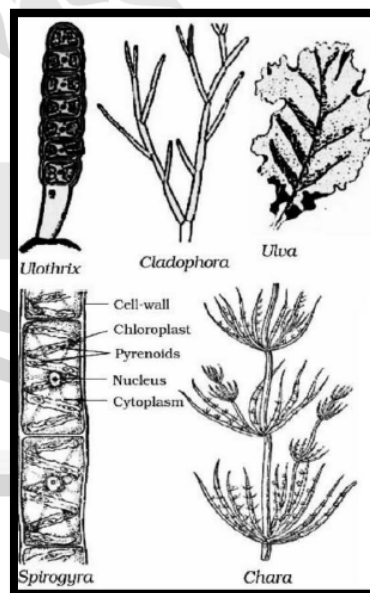
Algae are a diverse group of photosynthetic, eukaryotic microorganisms and multicellular organisms found primarily in aquatic environments.

- Mostly aquatic
- Body is simple (no root, stem, leaf)
- Study of Algae is called phycology
- Reproduce by spores or fragmentation

Examples: Spirogyra, Chlamydomonas

Algae		
Green Algae	Brown Algae	Red Algae
Chl 'a' × 'b'	Dominating Pigment	Dominating Pigment
Chlorella Chara Cladophora volvox Ulva Acetabularia	Pheucoxanthin Eg. Fucus Dyctyola Laminaria Kelp (Source of iodine)	Phycoerythrin Eg. Porphyra Gracilaria Geledium

CLASSES OF ALGAE AND THEIR MAIN CHARACTERISTICS						
Classes	Common Name	Major Pigments	Stored Food	Cell Wall	Flagellar Number and Position of Insertions	Habitat
Chlorophyceae	Green algae	Chlorophyll a, b	Starch	Cellulose and Pectose	2-8, equal, apical	Fresh water brackish water, salt water
Phaeophyceae	Brown algae	Chlorophyll a, c fucoxanthin	Mannitol, laminarin	Cellulose and algin	2,unequal lateral,	Fresh water (rare) brackish water, salt water
Rhodophyceae	Red algae	Chlorophyll a, d, phycoerythrin	Floridean starch	Cellulose and Pectin with Polysulphate ester	Absent	Fresh water (some), brackish water, salt water (most)



Bryophytes

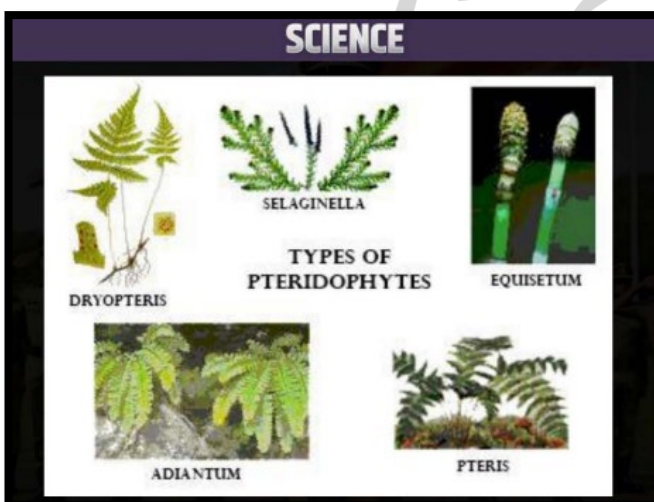
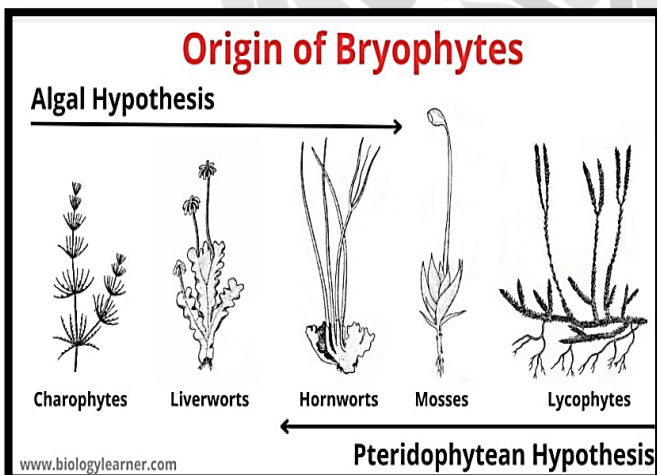
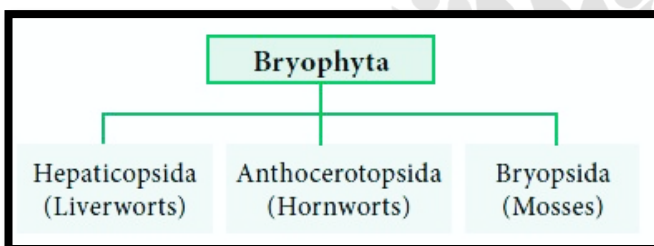
Bryophytes (mosses, liverworts, hornworts) are the earliest diverging land plants, likely originating 510–630 million years ago from green algae ancestors (Chlorophyceae).

- Amphibian of the plant kingdom.





- They can present on land but near water resource because water is necessary for their reproduction.
- Plant is made up of root, stem and leaflike structure.
- Vascular tissues are absent
- The term "Bryophyta" was proposed by "Robert Braun".
- The study of Bryophytes is known as Bryology.
- Hedwig is considered to be the father of Bryology. But according to some scientists it is believed that Cavers is the father of Bryology.
- Father of Indian Bryology is Prof. Shiv Ram Kashyap.



General characteristics:

1. Bryophytes are the first land plant. Amphibian of plant kingdom
 - It is believed that, they originated from on land through water. aquatic plant and they come
 - Because some bryophytes have characters similar to aquatic plants (eg. presence of air canal and formation of motile, flagellated male gametes.)
2. Bryophytes are known as amphibians of the plant kingdom, because these plants can live in soil but are dependent on water for fertilization.
3. Bryophytes are not considered as successful land plants because vascular tissue is absent and they need water for fertilization.
 - Due to the absence of vascular tissue bryophytes cannot grow very tall.
 - The process of water conduction in bryophytes takes place with the help of parenchyma.
 - Parenchyma is a living tissue.
4. Roots are absent in bryophytes. Stem like and leaf like structure of bryophytes are functionally similar to the stem and leaves of higher plants.
5. Bryophytes are sciophytes, i.e. bryophytes prefer to grow in moist (wet) and shady places

Pteridophytes

Pteridophytes are known as reptiles of plant kingdom:-

1. Pteridophytes are also called vascular cryptogames. Pteridophytes are vascular plants i.e. xylem and phloem are present in it.
 - In pteridophytes, vessels in xylem and companion cells in phloem are absent.
2. Pteridophytes are used for medicinal purposes and as soil binders. They are also frequently grown as ornamentals
3. Pteridophytes are more adapted terrestrial plants as compared to bryophytes. Because—
 - Vascular tissue is present in pteridophytes.
 - They have roots.
4. Pteridophytes are not completely successful terrestrial plants because they need water for





fertilization, so pteridophytes grow in cool, shady and moist places.

5. In pteridophyte, the plant body is completely differentiated into roots, stem and leaves.

- The primary root remains alive for short period.
- After some time, it is replaced by adventitious roots.
- Stem is erect or prostrate.
- When the stem is underground, it is known as rhizome.
- On the basis of leaves, pteridophytes are of two types
- First in which stem is smaller while leaves are larger.
- They are known as megaphyllous Pteridophytes. eg Ferns
- Second, in which stem is larger and leaves are smaller.
- They are called macrophyllous Pteridophytes. eg. Selaginella

Psilopsida : This is the class of most primitive vascular plants and their body is made of stem (rhizome), scaly leaves and rhizoids. eg. Psilotum A living fossil: **Lycopsidea**: Commonly known as club moss. Their body is made of root, stem and leaves (microphylls) Generally they produce cones/strobili at top of stem or branches. Cones are aggregation of sporophylls. eg. Lycopodium and Selaginella

Sphenopsida: Commonly known as horse tails. Their body is made of root, stem, rhizome and scaly leaves. Stem is distinctly differentiated into nodes and internodes. They also form cone/strobillus at top of aerial stem. eg Equisetum

Pteropsida : Commonly known as ferns, they are very good ornamental plants.

Their body is made of root, stem, rhizome large leaves (macrophylls or megaphylls) Leaves are circinate coiled in young stage.

Examples of Ferns:

- Azolla Aquatic fern (Smallest pteridophyte and biofertilizer)
- Adiantum (Maiden hair fern) (Walking fern This name is given to them
- due to rapid vegetative reproduction.

- Vegetative reproduction in Adiantum takes place by means of leaf tip.
- It spreads very fast.
- Alsophila - Tree fern (Largest pteridophyte)
- Pteridium
- Dryopteris
- Marsilea
- Pteris
- Salvinia
- In some examples of aquatic fern sporangia are found in specialised structure called sporocarp. eg. Marsilea, Azolla, Salvinia.
- Most of fossil fuels (coal) we obtained presently is the result of fossilization of giants ferns (Pteridophytes)

Gymnosperms

- The gymnosperms (gymnos = naked, sperma seed) and plants in which the ovule are not enclosed by any ovary wall and remain exposed, both before and after fertilization.
- Gymnosperms are very limited in distribution.
- They are mainly found in cold regions. In India gymnosperms are found on the slope of Himalayan mountains.
- As, they grow in slopes of mountains so they have to face water scarcity hence most of the gymnosperms are xerophyte.
- Study of gymnosperm known as gymnospermology. Lichen.
- Main plant body of gymnosperm is divided in to root, stem and leaves.
- The roots are generally tap roots. Note:- Roots in some genera have fungal association in the form of mycorrhiza (Pinus), while in some others (Cycas) small specialized roots called coralloid roots are associated with N₂ - fixing cyanobacteria.
- In Cycas roots are of two types i.e. tap root and coralloid roots.
- The stems are unbranched (Cycas) or branched (Pinus, Cedrus).
- The leaves may be simple or compound.
- In Cycas the pinnate leaves persist for a few years.





- The leaves in gymnosperms are well adapted to extreme conditions like temperature, humidity and wind.
- In conifers the needle like leaves reduce the surface area, their thick cuticle and sunken stomata also help to reduce water loss.
- Gymnosperm & Angiosperm are collectively included under spermatophyta i.e. seed bearing plants.
- Gymnosperms are naked seeded plant i.e. no fruit formation takes place in these plants.
- In gymnosperm embryo & seed formation take place but no fruit formation.
- All gymnosperms are vascular plants. Therefore vascular tissue presents i.e. xylem & phloem. Xylem lacks vessels & phloem lacks sieve tubes and companion cells.
- Note :- Exceptionally in xylem of Gnetum, Ephedra, Welwitschia true vessels are present.
- Early secondary growth takes place in gymnosperms stem, so it becomes woody.
- Most of the gymnosperms are arborescent (woody) and tree habit - but some are present as shrub. eg. Ephedra
- Some gymnosperm are liana or woody climbers. eg. Gnetum ula



Angiosperms

- Unlike the gymnosperms where the ovules are naked, in the angiosperms or flowering plants, the pollen grains and ovules are developed in specialized structures called flowers.
- In angiosperms, the seeds are enclosed in fruits.
- The angiosperms are an exceptionally large group of plants occurring in wide range of habitats.
- They range in size from the smallest. Wolffia to tall trees of Eucalyptus (over 100 metres). Highest anspatimay.
- They provideus with food, fodder, fuel, medicines and several other commercially important products.
- They are divided into two classes: the dicotyledons and the monocotyledons

	Monocot	vs	Dicot
Seed	 1 cotyledon		 2 cotyledons
Root	 Fibrous roots		 Tap roots
Flower	 Have petals in multiples of 3		 Have 4 or 5 petals
Leaf	 Narrow, parallel veins		 Oval or palmate, net-like veins
Vascular Bundles	 Scattered		 Ringed
Pollen Grains	 Have 1 pore or furrow		 Have 3 pores or furrows

- Roots, stems, and leaves modify to perform specialized functions like storage, support, and defense.
- Modifications include roots for storage (like carrots) or support (like prop roots on a banyan tree), stems for food storage (like ginger rhizomes) or photosynthesis (like in cactuses), and leaves for support (like tendrils on a pea plant), defense (like spines on a cactus), or trapping insects

